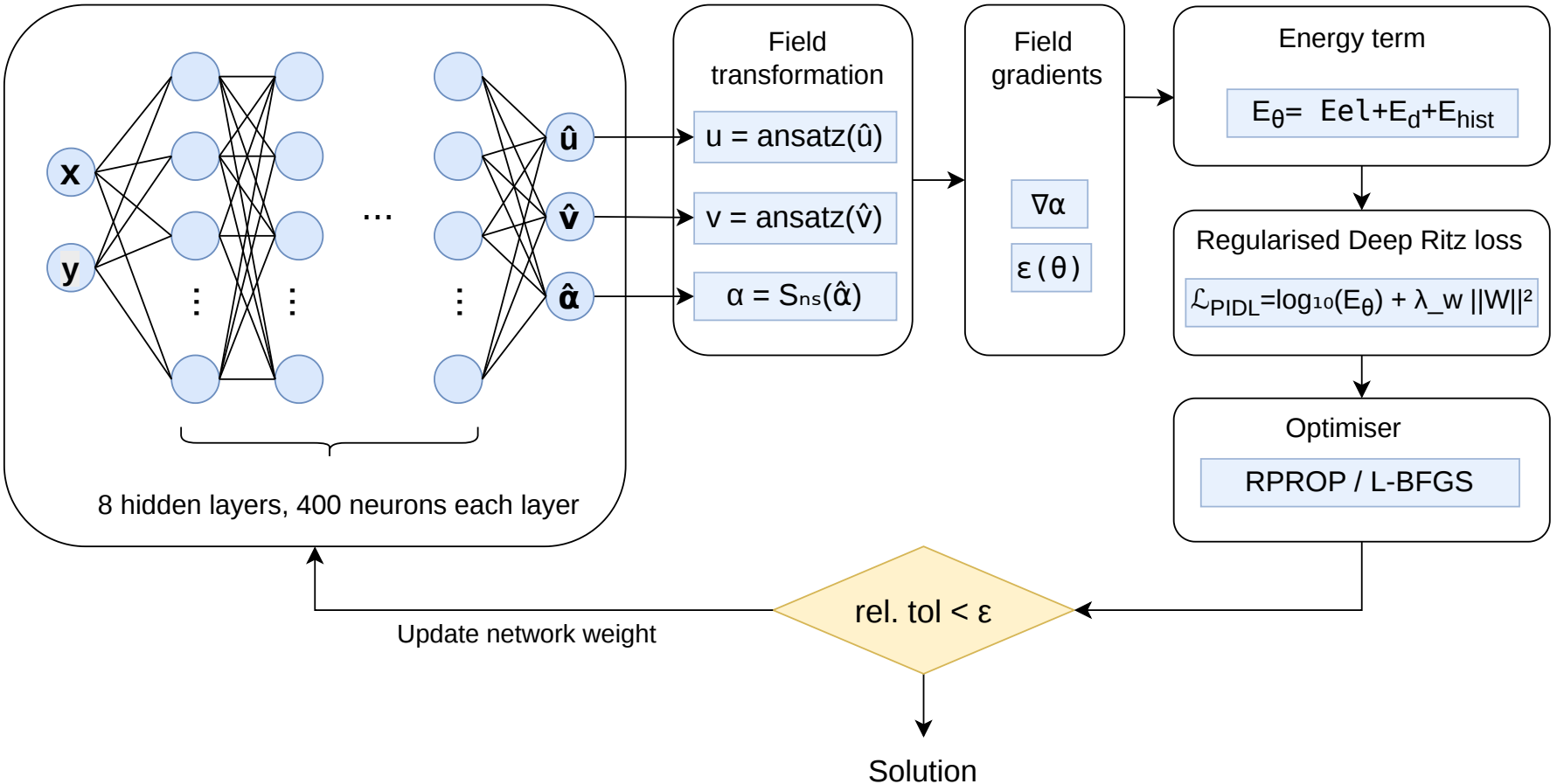
x
y
û
v̂
α̂
8 hidden layers, 400 neurons each layer
Field transformation
u = ansatz(û)
v = ansatz(v̂)
α = Sns(α̂)
Field gradients
∇α
ε(θ)
Energy term
Eθ= Eel+Ed+Ehist
Regularised Deep Ritz loss
ℒPIDL=log10(Eθ) + λ_w ||W||²
Optimiser
RPROP / L-BFGS
rel. tol < ε
Update network weight
Solution